TPC-418: A Finite-Family Shell-Parity Envelope

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Abstract

We prove a finite-family synthetic envelope for disjoint ordered complete prime shells. The crucial parity variable is the actual signed-block sign $\sigma_j = \epsilon_j(-1)^{n_j+1}$, where ϵ_j is only the global starting sign. Exact rational replay records this distinction, including a mixed-parity regression on which grouping by ϵ_j fails. Combining the envelope with the exact TPC417 endpoint-star/interior-bulk decomposition gives $\|Z\|_2 \leq 2/(a_{\min}\sqrt{H}) + 16B_*/V_-$. The result is finite and synthetic only; no growing, physical, or arithmetic conclusion is asserted.

1 Finite family and signs

Let $P_j = \{p : Q_j < p \leq 2Q_j\}$ be disjoint ordered complete shells, with $Q_j \geq 2$, $n_j = |P_j|$, $L \geq 2$, and all $p > N = 4H$. Set $\alpha_{j,r} = p_{j,r}^3/[Q_j^2(p_{j,r} - 1)]$. Global even indices are positive. The block start sign and actual block sign are

$$\epsilon_j = (-1)^{\sum_{\ell<j} n_\ell}, \qquad \sigma_j = \epsilon_j(-1)^{n_j+1}.$$

Define $b_j = \alpha_{j,n_j} - \alpha_{j,1}$ for even n_j and $b_j = \alpha_{j,n_j}$ for odd n_j, and group by σ_j: $B_\sigma = \sum_{\sigma_j=\sigma} b_j$, $B_* = \max(B_+, B_-)$.

2 Scalar envelope

For $f_Q(x) = x^3/[Q^2(x-1)]$, $f_Q'(x) = x^2(2x-3)/[Q^2(x-1)^2] > 0$. Also $\alpha > 1$. Since p is an odd prime and $2Q$ is even, $p \leq 2Q - 1$, so

$$\alpha \leq \frac{(2Q-1)^3}{Q^2(2Q-2)} < 4.$$

For increasing $x_1 < \cdots < x_n$, adjacent pairing shows the alternating sum is positive and at most x_n for odd n, and negative with absolute value at most $x_n - x_1$ for even n. Thus $|A| \leq B_*$. Even shells have $b_j < 3$, odd shells $b_j < 4$, and odd-shell signs alternate, yielding

$$B_* < 3E + 4\lceil O/2 \rceil \leq 3K + 1.$$

3 Exact matrix bound

With $T_d = H^2/(H^2 + d^2)$ and $S_r = \sum_{s\neq r} T_{|s-r|}^2$, exact CRT masks and diagonal deletion give

$$M_{0r} = P_- T_r, \quad M_{rs} = -AT_{r-s}, \quad D_0 = V_- S_0, \quad D_r = V_- S_r + V_+(S_r - T_r^2).$$

Therefore $Z = D^{-1/2}MD^{-1/2} = \left[\begin{smallmatrix} 0 & q^T \\ q & C \end{smallmatrix}\right]$. Since $S_r \geq H/4$, $P_-^2 \leq m_- V_-$, and $V_- \geq m_- a_{\min}^2$, $\|q\|_2 \leq 2/(a_{\min}\sqrt{H})$. The one-sided kernel sum is at most $2H$, while $D_r \geq V_- H/4$; hence symmetry gives $\|C\|_2 \leq 16B_*/V_-$. Consequently

$$\boxed{\|Z\|_2 \leq \frac{2}{a_{\min}\sqrt{H}} + \frac{16B_*}{V_-} < \frac{2}{\sqrt{H}} + \frac{16(3K+1)}{m_-}}.$$

4 Audit and scope

The exact certificate contains a fixed four-shell replay, a small complete multi-shell replay, and a mixed odd/even regression. Producer and independent checker use exact rational strings; mutation tests cover the old sign grouping. This is a finite synthetic envelope, not a growing uniform theorem, and it does not identify physical h_0, prove arithmetic signs or L^2 savings, pay fixed-power credit, close Route-B, or imply twin primes.